

1. Explain the difference between a has-a and is-a relationship among classes.
A has-a relationship means a class contains another class as a field
An is-a relationship means a class inherits from another class
2. If a base class has a public method go() and a derived class has a public method stop(), which methods will be available to an object of the derived class?
Both methods would be available
3. Compare and contrast implementing an abstract method to overriding a method.
Implementing gives code for an abstract method, while overriding changes an existing method.
4. Compare and contrast an abstract method to overriding a method.
An abstract class cannot be instantiated and is supposed to be inherited. Interface is only methods, classes implement it.
6. Use the following classes to answer the questions below:
 - a) What type of method is doThat() in Wo?
It's an abstract method because it has no code and must be provided by any class that implements the interface.
 - b) What is Wo?
Wo is an interface, which means it defines methods that other classes must implement.
 - c) Why is doThat() implemented in Roo?
Since Roo implements the Wo interface, it must provide code for all the methods defined in that interface.
 - d) List the methods available to a Roo object.
A Roo object has doThis() from Roo, doNow() from Bo, and doThat() from Roo due to the interface. But, doThat() in Roo is private, so it exists in the object but cannot be accessed from outside the class.
 - e) How does the implementation of doThis() in Roo affect the implementation of doThis() in Bo?
Roo overrides Bo's doThis() method, so when called on a Roo object, Roo's version is used instead of Bo's
 - f) What action does the statement super(1) in Roo perform?
It calls Bo's constructor with the value 1, which sets Bo's x variable.
 - g) Can the doThis() method in Bo be called from a Roo object? If so, how?
Inside Roo, it can be called using super.doThis(), but from outside the class it cannot be called directly because Roo's version overrides it.
 - h) Can a method in Roo call the doThis() method in Bo? If so, how?
Yes, a method in Roo can call Bo's version by using super.doThis()

